Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE/ *NASIONALE SENIOR SERTIFIKAAT*

GRADE/*GRAAD* 10

NOVEMBER 2018

TECHNICAL SCIENCES P2/*TEGNIESE WETENSKAPPE V2*
MARKING GUIDELINE/*NASIENRIGLYN*

MARKS/*PUNTE*: 150

This marking guideline consists of pages 7./
Hierdie nasienriglyn bestaan uit 7 bladsye.

QUESTION/*VRAAG* 1

1.1 A ✓✓ (2)
1.2 B ✓✓ (2)
1.3 C ✓✓ (2)
1.4 B ✓✓ (2)
1.5 D ✓✓ (2)
1.6 B ✓✓ (2)
1.7 B ✓✓ (2)
1.8 D ✓✓ (2)
1.9 C ✓✓ (2)
1.10 A ✓✓ (2)
[20]

QUESTION/*VRAAG* 2

2.1 Low density,✓ Unreactive ✓ (2)
Lae digtheid, ✓ *onreaktief* ✓

2.2 2.2.1 Isotopes ✓✓*Isotope* ✓✓ (2)

2.2.2

	Electrons e^- **Elektrone e^-**	**Protons p^+** **Protone p^+**	**Neutrons n^0** **Neutrone n^0**
$_1{}^3$**H**	1✓	1	2✓
$_1{}^2$**H**	1✓	1	1✓
$_2{}^4$**He**	2✓	1	2✓

(6)

2.3 2.3.1 Noble gases ✓✓or Inert gases✓✓
Edelgasse ✓✓ *of onaktief gasse* ✓✓ (2)

2.3.2 $1s^2 2s^2 2p^6 3s^2 3p^6$ ✓✓ (2)
[14]

QUESTION/*VRAAG* 3

3.1 3.1.1 Copper Carbonate ✓ Koperkarbonaat ✓ (1)

3.1.2 Sugar water ✓ *Suikerwater* ✓ (1)

3.1.3 Barium nitrate ✓ *Bariumnitraat* ✓ (1)

3.1.4 Carbon dioxide ✓ *Koolstofdioksied* ✓ (1)

3.1.5 Silver iodide ✓ *Silwer jodium* ✓ (1)

3.1.6 Salt water ✓ *Soutwater* ✓ (1)

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3.2 3.2.1 KCl ✓ (1)

3.2.2 Electrons are transferred√ from one atom (metal) ✓ to another (non-metal) ✓.
This forms positive and negative ions which attract each other.
Elektrone word oorgedra ✓ van een atoom (metaal) ✓ na 'n ander (nie-metaal) ✓. Dit vorm positiewe en negatiewe ione wat mekaar aantrek (3)

3.2.3 Dissociation ✓ (Accept: Ionisation ✓) Dissosiasie ✓ (Aanvaar: Ionisation ✓) (1)

3.2.4 $KCl(s)$ ✓ $\rightarrow K^+(aq)$ ✓ + $Cl^-(aq)$ ✓
Or/*Of*
$KCl(s)$ ✓ $\rightarrow KOH(aq)$ ✓ + $HCl(aq)$ ✓ (3)
[14]

QUESTION/*VRAAG* 4

4.1 A – Ammeter ✓✓ *Ammeter* ✓✓
B – Battery ✓✓ or Cells ✓✓ *Battery of Selle* ✓✓
C – Switch ✓✓ *Skakelaar* ✓✓ (6)

4.2 It must form ions in solution ✓✓ that are free to move./
Dit moet ione in oplossing vorm ✓✓ *wat vry is om te beweeg*. (2)

4.3

Name of the substance ***Naam van stof***	**Conduct or not: ✓ or X** ***Leiding of nie*** **(✓ or ×)**
Potassium Chloride Crystal *Kaliumchloriedkristal*	**X**
Distilled water *Gedistilleerde water*	**X**
Potassium Chloride solution *Kaliumchloried oplossing*	✓

(3)

4.4 Increase in the reading ✓✓ on apparatus **A** as a result of an increase in the concentration of solution.
*Toename in die lesing ✓✓ op apparaat **A** as gevolg van 'n toename in die konsentrasie van oplossing.*

4.5 Dependent variable- Conductivity ✓ or Ammeter reading ✓ or glowing (brightness of bulb) ✓
Independent variable – type of material used ✓ (to test conductivity)
Control variable- Number of cells (battery) ✓ or type of cells (battery) ✓
Afhanklike veranderlike – Konduktiwiteit ✓ of Ammeterlesing ✓ of gloeiend (gloei van gloeilamp) ✓
Onafhanklike veranderlike – tipe materiaal wat gebruik word ✓ (om geleidingsvermoë te toets)
Beheer veranderlike – Aantal selle (battery) ✓ of tipe selle (battery) ✓ (3)

4.6 Light bulb, ✓ the bulb will glow brighter ✓✓ to show conductivity./
Gloeilamp, ✓ *die gloeilamp sal helderder gloei* ✓✓ *om geleidingsvermoë te toon.*

4.7 4.7.1 Non-magnetic ✓/ *Nie-magneties* ✓ (1)

4.7.2 Magnetic ✓/ *Magneties* ✓ (1)

4.7.3 Non-magnetic ✓/ *Nie-magneties* ✓ (1)

4.8 4.8.1 It is a material that does not allow heat energy to pass through it. ✓✓
Dit is 'n materiaal wat nie toelaat dat hitte-energie daardeur beweeg nie. ✓✓ (2)

4.8.2 Plastic, ✓ Wool, ✓ Fibreglass, ✓ Polystyrene ✓ etc. (Any 2)
Plastiek, ✓ *Wol*, ✓ *Veselglas*, ✓ *Polistireen* ✓ *ens. (Enige 2)* (2)

4.9 In ringing bells ✓ / *In klokke wat lui* ✓
In floppy disks ✓ *I in diskette* ✓
In alarm systems ✓ / *In alarmstelsels* ✓
In electric motors ✓ / *In elektriese motors* ✓
Electric cranes with magnets ✓ / *Elektriese hyskrane met magnete* ✓
In sound systems ✓ *In klankstelsels* ✓ (Any 3 / *Enige 3*) (3)
[29]

QUESTION/*VRAAG* 5

5.1 A cation is a positive ion ✓ (atom or molecule)
An anion is a negative ion ✓ (atom or molecule)/
'n Katioon is 'n positiewe ioon ✓ *(atom of molekule)*
'n Anioon is 'n negatiewe ioon ✓ *(atoom of molekule)* (2)

5.2 5.2.1 Potassium ✓ +1 ✓ *Kalium* ✓ *+1* ✓ (2)

5.2.2 Aluminium ✓ +3 ✓ *Aluminium* ✓ *+ 3* ✓ (2)

5.2.3 Hydroxide ✓ -1 ✓ *hidroksied* ✓ *-1* ✓

5.3 5.3.1 Iron (II) oxide ✓✓ Yster (II) oksied ✓✓ (2)

5.3.2 Calcium (II) Chloride ✓✓ *Kalsium (II) Chloried* ✓✓ (2)

5.4 5.4.1 HBr ✓✓ (2)

5.4.2 CuO ✓✓ (2)

5.4.3 $MgCl_2$ ✓✓ (2)

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5.5 5.5.1 Na_2 is incorrect ✓ rather 2Na as Sodium is not a diatomic molecule it is a metal in group 1. ✓
$(NaOH)_2$is incorrect√ it must be NaOH as Na^+OH^- is an ionic compound ratio is 1 : 1. ✓/
Na_2 is inkorrek ✓ eerder 2Na aangesien natrium nie 'n diatomiese molekule is nie, dit is 'n metaal in groep 1. ✓
$(NaOH)_2$ is inkorrek ✓ dit moet NaOH wees as Na + OH- is 'n ioniese verbinding met 'n verhouding van 1 : 1. ✓ (4)

5.5.2 Mg_2Cl_2 is incorrect ✓ it must be $MgCl_2$. ✓ Magnesium is a 2+ ion and Chlorine is 1- ion so the ratio is 2 : 1. ✓/
Mg_2Cl_2 is inkorrek ✓dit moet MgCl2 wees. Magnesium is 'n 2+ ioon en Chloor is 1- ioon dus die verhouding is 2 : 1 ✓. (3)
[25]

QUESTION/*VRAAG* 6

6.1 $CaCO_3(s)$ ✓ → $CaO(s)$ ✓ + $CO_2(g)$ ✓ (3)

6.2 Reactant mass *Reaktante massa* = {40 + 3(16)} ✓ = 100 amu/g.mol^{-1} ✓
Products mass *Produkte massa* = { (40 + 16) + (12 + 2(16)) } ✓ = 100 amu/ g.mol^{-1} ✓
Reactants mass/*Reaktante massa* = Product mass/*Produkte massa* (4)

6.3 <u>Heat is needed</u> ✓✓ to break up $CaCO_3(s)$ into simple compounds.
<u>*Hitte is nodig*</u>*. ✓✓ Om $CaCO_3$ (s) op te los in eenvoudige verbindings*. (2)

6.4 The gas is passed through lime water ✓ that is clear and a <u>milky colour</u> ✓✓ will be observed.
Die gas word deur kalkwater oorgedra. ✓ *Die water helder en 'n melkagtige kleur* sal ✓✓ *waargeneem word.* (3)

6.5 Yes ✓ To mix the substances evenly ✓ *Om die stowwe eweredig te meng* ✓
OR To distribute the heat evenly ✓ ***OF*** *Om die hitte eweredig te versprei* ✓
To prevent the cement from settling and hardening ✓ *Om te verhoed dat die sement verhard* ✓ (2)
[14]

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QUESTION/*VRAAG 7*

7.1 7.1.1 Halogens ✓✓/ *Halogene* ✓✓ (2)

7.1.2

	^{35}Cl	^{37}Cl
Number of protons *Aantal protone*	17	17
Number of electrons *Aantal elektrone*	a. 17 ✓	b. 17 ✓
Number of neutrons *Aantal neutrone*	c. 18 ✓	d. 20 ✓
Number of Nucleons *Aantal nukleone*	35	37

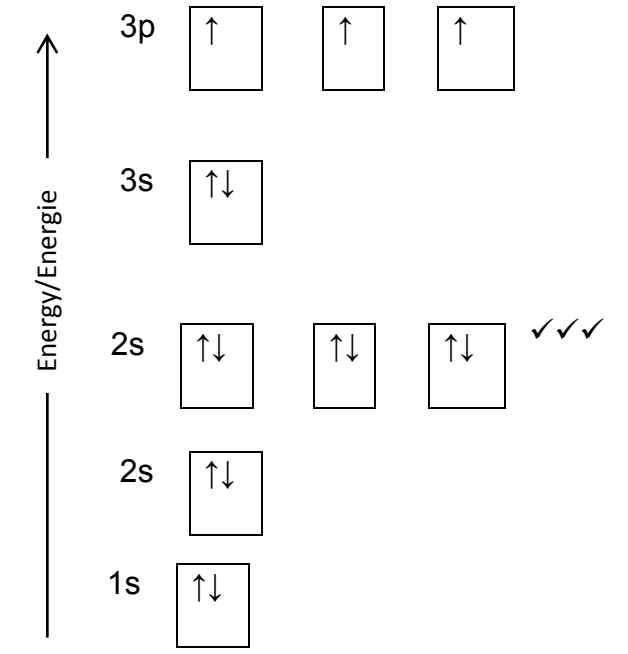
(4)

7.2 Nitrogen; ✓ $_7{}^{15}N$ ✓✓ / Stikstof ✓ $_7{}^{15}N$ ✓✓ (3)

7.3 7.3.1 True ✓ *Waar* ✓ (1)

7.3.2 False ✓ Onwaar ✓ (4)

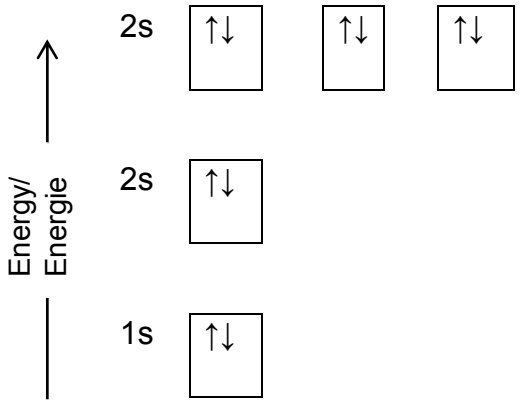
7.4 7.4.1



✓✓✓

(3)

7.4.2



(2)

[16]

QUESTION/*VRAAG* 8

8.1 Heat is a form of energy (that can give rise to temperature if transferred). ✓✓
Temperature is a measure of how hot or cold an object/body is. ✓✓/
Hitte is 'n vorm van energie (dit kan aanleiding gee tot temperatuur indien dit oorgedra word). ✓✓
Temperatuur is 'n mate van hoe warm of koud 'n voorwerp / liggaam is. ✓✓ (4)

8.2 Alcohol thermometer ✓ / *Alkohol termometer* ✓
Mercury thermometer ✓ / *Kwiktermometer* ✓
Thermoelectric thermometer ✓ / *Termo-elektrisiteit* ✓ (3)

8.3 It is used in medicine. ✓ / *Dit word in medisyne gebruik*. ✓
It is used in scientific research ✓/ science laboratories ✓
Dit word gebruik in wetenskaplike navorsing ✓ / *wetenskaplaboratoriums* ✓
It is used in meteorology ✓/ study of weather ✓
Dit word gebruik in meteorologie ✓ / *studie van weer* ✓
To control and regulate processes in industry ✓ /
Om prosesse in die bedryf te beheer en te reguleer ✓ (Any 3/*Enige 3*) (3)

8.4 8.4.1 A – Thermometer ✓/ *Termometer* ✓
B – Glass rod ✓/ *Glasstang* ✓
C – Test Tube ✓ / *Proefbuis* ✓
D – Bunsen burner ✓ / *Bunsenbrander* ✓
E – Glass beaker ✓ / *Glasbeker* ✓ (5)

8.4.2 It is used as a stirrer ✓✓ so that heat is evenly distributed./
Dit word gebruik om in mengsel te roer ✓✓ *sodat hitte eweredig versprei word.* (2)

8.4.3 It is flammable. ✓ *Dit is vlambaar*. ✓ (1)

8.4 Sound waves/Seismic waves/Earthquake (Any 2) ✓✓/
Klankgolwe/Seismiese golwe/Aardbewings (Enige 2) ✓✓ (2)
[18]

TOTAL/*TOTAAL*: 150

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